

Rachit Sabharwal

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Professional Summary

Ph.D. Candidate (expected Aug 2026) with 6+ years building biomedical AI systems and generating real-world evidence from EHR, clinical-trial, and population-cohort data across biopharma and academic-medical-center environments. Dissertation develops explainable multimodal AI for severe TBI outcome prediction, fusing imaging, physiologic, and structured EHR signals. Shipped production analytics and decision-support tools at Dow, Genentech, and BMS; track record of translating research into peer-reviewed publications and deployable systems.

Education

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| Ph.D. The University of Texas Health Science Center at Houston, Epidemiology and Biostatistics | Houston, TX
Expected Aug 2026 |
| <ul style="list-style-type: none">• Focus: Explainable AI• Certificates: Advanced Data Science | |
| Cert. The University of California, Berkeley, Software Development and Programming | Berkeley, CA
Apr 2025 |
| M.S. The University of Texas Health Science Center at Houston, Biostatistics | Houston, TX
May 2022 |
| B.S. University of Rochester, Environmental Health | Rochester, NY
May 2018 |

Experience

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| The University of Texas Health Science Center at Houston
Research Data Scientist | Houston, TX
Feb 2020 – present |
| <ul style="list-style-type: none">• Developing an XAI multimodal biomedical AI model fusing heterogeneous clinical signals (imaging, physiologic, structured EHR) for TBI outcome prediction; benchmarked unimodal deep learning against classical-ML baselines across multiple biomedical data sources.• Designed the multi-site star schema and ETL for the R33 phase of DETECT-RPC, supervising 2 M.S. students while harmonizing clinical records from 6 academic medical centers (UCSF, UAB, UTSW, JHU, UTH, BCM).• Led a team evaluating LLM feasibility for automated extraction and classification of free-text physician notes in DETECT-RPC, informing the project's clinical NLP pipeline design.• Cut data-munging turnaround 24x by re-architecting the TexasCARES ETL pipelines across multiple data sources; stood up CI/CD automated reporting that ingests new data weekly, runs quality checks, and publishes stakeholder outputs without manual intervention.• Co-authored 4 peer-reviewed publications using survival analysis and linear mixed models on population-based cohorts (vaccine non-response, pediatric COVID comorbidities). | |
| The Dow Chemical Company
ML R&D Intern | Lake Jackson, TX
May 2025 – Aug 2025 |
| <ul style="list-style-type: none">• Benchmarked Bayesian statistical models (Gaussian process regression, Bayesian neural networks, Monte Carlo dropout) against frequentist counterparts across 25+ measurement-error scenarios using UQ calibration metrics.• Authored an internal Bayesian UQ white paper (MiLOS) adopted as organizational guidance for future predictive modeling projects; shipped a decision-support tool estimated at \$15M annual savings. | |

Genentech

Informatics R&D Intern

San Francisco, CA
May 2021 – Jan 2022

- Reproduced the published GLIT multimodal GNN for drug-induced liver injury (DILI) prediction, then designed and validated an alternative MLP combining ECFP molecular fingerprints with drug diffusion profiles that matched GLIT within 10% predictive performance at reduced model complexity.
- Contributed to an internal Neo4j knowledge graph of drug-gene interactions used by R&D teams for therapeutic signature work.

Procter & Gamble

Advanced Analytics Co-Op

Cincinnati, OH
Jan 2021 – May 2021

- Delivered predictive and explanatory models on consumer behavior; raised analyst throughput 1.5x with GCP big-data ETL pipelines using Apache Airflow.

Bristol Myers Squibb

Data Engineering Intern

San Francisco, CA
June 2020 – Aug 2020

- Built PatRec, a full-stack biomedical patent recommendation system chaining chemical-entity NER, NCBI federated literature search, and TF-IDF cosine-similarity ranking to surface relevant patents from an 80M-patent corpus for internal R&D scientists.

Selected Publications

Trust and Uncertainty Quantification in Machine Learning Models Under Measurement Error

Aug 2025

Sabharwal R

(The Dow Chemical Company, Internal White Paper)

Baseline characteristics of SARS-CoV-2 vaccine non-responders in a large population-based sample

May 2024

Yaseen A, DeSantis SM, **Sabharwal R**, et al.

10.1371/journal.pone.0303420 (PLoS One)

Factors associated with elevated SARS-CoV-2 immune response in children and adolescents

Aug 2024

Messiah SE, Abbas R, **Sabharwal R**, et al.

10.3389/fped.2024.1393321 (Frontiers in Pediatrics)

Skills

Biomedical AI & Statistical Modeling: Multimodal Deep Learning, Graph Neural Networks, Explainable AI, Bayesian Inference & UQ, Causal Inference, Survival Analysis, Mixed Models, Real-World Evidence, NLP, LLMs, Biomedical Data Integration

ML/AI Libraries: PyTorch, PyTorch Lightning, Hugging Face, TensorFlow, TensorFlow Probability, scikit-learn, tidymodels, brms, Optuna, SHAP, RDKit

Programming & Algorithms: Python, R, SQL, Bash

MLOps & Cloud: Docker, Kubernetes, Apache Airflow, GitHub/GitLab CI/CD, GCP, AWS, Azure, PostgreSQL, pgvector, Neo4j, BigQuery